README + AI Usage Report

Travel Exploration Local

Milestone 3 – Frontend, Database, and External API Integration

Project Overview

Travel Exploration Local is a full-stack web application designed to help users discover and store location-based points of interest, referred to as “gems.” These gems include places such as parks, restaurants, and museums. The system allows users to create, view, update, and delete gem records while storing all data in a PostgreSQL database hosted on Neon.

The application integrates a React frontend with a Spring Boot backend API and uses the Google Maps JavaScript API to visually display stored locations.

🧩 Task

1. Frontend (ReactJS)

- Implement a React app with **three required pages**:
    1. Home/Dashboard – overview and navigation.
    2. Data Display – retrieves data via backend GET endpoint.
    3. Form Page – posts data via backend POST endpoint.
- Use React Router and state management (useState or useReducer).
- Include a responsive layout and consistent styling.
- Connect all pages to API calls with Axios or fetch().

⚙️ All listed features must be implemented even if only a subset is graded now. Later milestones depend on this codebase.

2. Database Integration

- Configure MySQL, PostgreSQL, or H2 database.
- Use Spring Data JPA to create Entities and Repositories.

- Replace temporary storage with persistent tables.
- Implement relationships (1-Many, Many-Many) and input validation.

3. External API Integration

- Choose a public API relevant to your app.
- Retrieve and store selected data on your backend.
- Display integrated API data in React.
- Handle errors and rate limits gracefully.

4. Authentication & Security

- Add user registration and login with Spring Security.
- Protect routes using JWT or session-based authorization.
- Restrict specific endpoints to authenticated users.

5. Documentation

- Database schema diagram (ERD or JPA Relationships)
- Updated endpoint table with auth requirements
- Example API responses + **AI Usage Log**

💡 To Succeed

- Demonstrate secure data flow from frontend → backend → DB.
- Follow naming and structure best practices.
- Test CRUD and login before demo.

Demo Video

https://drive.google.com/file/d/1rcylihWxo3ZjbXEMRhQC8MDVL4Bw4a6g/view?usp=sharing

Tool Used

ChatGPT

Purpose of AI Use

AI was used to assist with:

- Correcting API endpoint mismatches
- Implementing Google Maps integration
- Generating documentation and structuring project reports

Prompts Used

Examples of prompts used include:

- “Where do I display integrated API data in React page”
- “Show all pages with correct backend connections”
- “Create README with AI log and separate documentation”